Experiment 1: Crafting Zero-Shot, One-Shot, and Few-Shot Prompts

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Aim

To understand the basic concepts of Zero-Shot, One-Shot, and Few-Shot prompting techniques and observe how different prompting methods influence AI-generated responses.

Objective

After completing this experiment, students will be able to:

- Understand the concept of Prompt Engineering.
- Differentiate between Zero-Shot, One-Shot, and Few-Shot prompting.
- Use Generative AI tools to generate responses using different prompt styles.
- Compare AI responses based on clarity, accuracy, and completeness.
- Identify which prompting technique produces better results for different tasks.

Tools Required

- ChatGPT – https://chatgpt.com
- Google Gemini – https://gemini.google.com *(Optional)*
- Internet Connection
- Web Browser (Google Chrome / Microsoft Edge)

General Procedure

Step 1: Open a web browser and visit **https://chatgpt.com**.
Step 2: Sign in using your Google, Microsoft, or email account.
Step 3: Enter the given prompt into the prompt/input box.
Step 4: Click **Send** or press **Enter** to generate the response.
Step 5: Observe the generated output, compare the responses, and record your observations.

Introduction

Prompt Engineering is the process of designing clear and effective instructions (prompts) to obtain the desired output from an AI model.
The three basic prompting techniques are:
Zero-Shot Prompting
The AI is asked to perform a task **without providing any examples**.
One-Shot Prompting
The AI is given **one example** before asking it to perform the task.
Few-Shot Prompting
The AI is given **multiple examples (two or more)** before asking it to perform the task.
These techniques help improve the quality, consistency, and accuracy of AI-generated responses.

Laboratory Task L1.1
Task Name
Generate a Response Using a Zero-Shot Prompt

Tool Used

- ChatGPT

Sample Prompt
Explain Machine Learning in simple words suitable for a first-year B.Tech student.

Procedure

1. Open Google Chrome or Microsoft Edge.
2. Visit **https://chatgpt.com**.
3. Sign in using your account.
4. Paste the sample prompt into the chat box.
5. Click **Send** and read the generated response.

Expected Output
The AI should explain Machine Learning in simple language without requiring any example from the user.
The response should include:

- Definition of Machine Learning
- Simple explanation
- One or two real-life examples

Actual Output
Paste or attach the AI-generated explanation here

Laboratory Task L1.2
Task Name
Generate a Response Using a One-Shot Prompt

Tool Used

- ChatGPT

Sample Prompt
Example:
Input: Apple
Output: Apple is a healthy fruit rich in vitamins.
Now generate a similar sentence for:Banana

Procedure

1. Open ChatGPT.
2. Sign in using your account.
3. Copy and paste the sample prompt.
4. Click **Send**.
5. Observe the generated response.

Expected Output
The AI should generate a similar sentence following the given example.
Example:
Banana is a nutritious fruit that provides energy and essential nutrients.

Actual Output
Paste or attach the generated response here.

Laboratory Task L1.3
Task Name
Generate a Response Using a Few-Shot Prompt

Tool Used

- ChatGPT

Sample Prompt
Examples:
Apple → Apple is a healthy fruit rich in vitamins.
Mango → Mango is a sweet tropical fruit enjoyed during summer.
Orange → Orange is a citrus fruit that is rich in Vitamin C.
Now generate a similar sentence for:
Guava

Procedure

1. Open ChatGPT.
2. Sign in using your account.
3. Enter the sample prompt into the chat box.
4. Click **Send**.
5. Read the generated response carefully.

Expected Output
The AI should generate a sentence similar to the previous examples.
Example:
Guava is a nutritious fruit rich in Vitamin C and dietary fiber.
Students should observe that providing multiple examples produces more consistent responses.

Actual Output
Paste or attach the AI-generated response here.

Laboratory Task L1.4
Task Name
Compare Zero-Shot, One-Shot, and Few-Shot Responses

Tool Used

- ChatGPT

Task
Compare the responses generated in Tasks **L2.1**, **L2.2**, and **L2.3**.

Procedure

1. Review the responses generated in the previous tasks.

2. Observe the quality of each response.
3. Compare the responses using the given criteria.
4. Record your observations.
5. Discuss which prompting technique produced the best result.

Comparison Criteria

- Clarity
- Accuracy
- Completeness
- Ease of Understanding
- Response Quality

Expected Output

Students should identify differences such as:

- Zero-Shot provides a general response.
- One-Shot follows the given example.
- Few-Shot produces a more consistent and accurate response.

Actual Output

Write your observations.

Prompt Type	Observation
Zero-Shot	
One-Shot	
Few-Shot	

Laboratory Task L1.5

Task Name

Prepare a Comparison Table (give your own prompt basing comparison table)

Tool Used

- ChatGPT

Procedure

1. Review all previous outputs.
2. Compare the three prompting techniques.
3. Complete the following table.
4. Write your conclusion.
5. Submit the completed observation sheet.

Comparison Table

Feature	Zero-Shot	One-Shot	Few-Shot
Prompt contains examples	No	One	Three
Easy to create			
Response clarity			
Response accuracy			

Completeness			
Consistency			
Suitable for beginners			
Best for complex tasks			
Overall Rating (1–5)			

Student Conclusion

Answer the following questions.

1. Which prompting technique produced the best response?

2. Which prompting technique was easiest to understand?

3. Which prompting technique would you use for complex tasks? Why?

4. What did you learn from this experiment?

Expected Output

Students should complete the comparison table and conclude that:

- **Zero-Shot** is useful for simple tasks where no examples are needed.
- **One-Shot** helps the AI understand the expected format through one example.
- **Few-Shot** generally produces more accurate, consistent, and context-aware responses because it learns from multiple examples.

Actual Output

Students should attach:

- The completed comparison table.
- Written observations.
- Final conclusion.

Observation Table

Task No.	AI Tool Used	Type of Activity	Completed (✓/✗)
L2.1	ChatGPT	Zero-Shot Prompting	
L2.2	ChatGPT	One-Shot Prompting	
L2.3	ChatGPT	Few-Shot Prompting	
L2.4	ChatGPT	Response Comparison	
L2.5	ChatGPT	Comparison Table & Conclusion	

Result

The experiment was successfully completed. Students explored the three basic prompting techniques—**Zero-Shot, One-Shot, and Few-Shot Prompting**—using the same AI tool. By comparing the generated responses, they observed how providing examples influences the quality, clarity, and completeness of AI-generated outputs. The experiment demonstrated the importance of prompt engineering in obtaining accurate and useful responses from Generative AI systems.

Learning Outcomes

After completing this experiment, students will be able to:

1. Define Prompt Engineering and explain its importance.
2. Differentiate between Zero-Shot, One-Shot, and Few-Shot prompting techniques.

3. Write effective prompts for AI-based tasks.
4. Compare AI-generated responses using simple evaluation criteria.
5. Identify suitable prompting techniques for different problem-solving scenarios.
6. Improve the quality of AI responses by selecting appropriate prompting methods.
7. Develop confidence in using Generative AI tools for academic and everyday tasks.

Viva-Voce Questions

1. What is Prompt Engineering?
2. What is a prompt?
3. Define Zero-Shot Prompting.
4. Define One-Shot Prompting.
5. Define Few-Shot Prompting.
6. Which prompting technique provides one example?
7. Which prompting technique provides multiple examples?
8. Why is Few-Shot Prompting often more effective than Zero-Shot Prompting?
9. Name two Generative AI tools that support prompt engineering.
10. Which prompting technique would you choose for a complex task? Why?

Assignment

Topic: Exploring Prompt Engineering Techniques

1. Write one **Zero-Shot**, one **One-Shot**, and one **Few-Shot** prompt for the task **"Generate a short paragraph about Renewable Energy."**
2. Execute all three prompts using **ChatGPT or Gemini**.
3. Compare the outputs based on:
    - Clarity
    - Accuracy
    - Completeness
4. Write a **100–150 word conclusion** explaining which prompting technique produced the best result and why.
5.

Experiment 2: Advanced Prompt Optimization Exercise

Aim

To understand how prompt optimization techniques improve the quality, accuracy, and relevance of AI-generated responses.

Objective

After completing this experiment, students will be able to:

- Understand the importance of writing effective prompts.
- Learn how context, constraints, and output formatting improve AI responses.
- Compare vague prompts with optimized prompts.
- Analyze improvements in AI-generated outputs.
- Apply prompt optimization techniques for academic and technical tasks.

Tools Required

- ChatGPT (https://chatgpt.com)

- Google Gemini (Optional)
- Internet Connection
- Web Browser (Google Chrome / Microsoft Edge)

General Procedure
Step 1: Open Google Chrome or Microsoft Edge.
Step 2: Visit **https://chatgpt.com** and sign in using your Google, Microsoft, or Email account.
Step 3: Enter the given prompt in the prompt box.
Step 4: Click **Send** to generate the response.
Step 5: Observe the output, compare different prompts, and record your observations.

Introduction
Prompt Optimization is the process of improving a prompt by adding **clear instructions, context, constraints, and desired output formats**. Better prompts help AI generate more accurate, relevant, and useful responses.
A good prompt usually includes:

- Clear task
- Context
- Target audience
- Output format
- Constraints (length, style, etc.)

For example:
Vague Prompt
Explain Python.
Optimized Prompt
Explain Python programming in simple language for first-year CSE students. Include its definition, three features, three applications, and one simple example in bullet points.
The optimized prompt provides better guidance, resulting in a more structured and useful response.

Laboratory Task L2.1
Task Name
Write a Vague Prompt and Observe the AI Response

Tool Used

- ChatGPT

Sample Prompt
Explain Artificial Intelligence.

Procedure

1. Open ChatGPT in your browser.
2. Sign in using your account.
3. Enter the sample prompt.
4. Click **Send**.
5. Observe the generated response.

Expected Output
The AI provides a general explanation of Artificial Intelligence. The response may not include enough detail.
Example:

- Definition of AI
- Brief explanation
- One or two examples

Actual Output
Paste the AI-generated response here.

Laboratory Task L2.2
Task Name
Improve the Prompt by Adding Context

Tool Used

- ChatGPT

Sample Prompt
Explain Artificial Intelligence in simple language for first-year BTech students. Include its definition, importance, and three real-world applications.

Procedure

1. Open ChatGPT.
2. Enter the improved prompt.
3. Click **Send**.
4. Read the generated response.
5. Compare it with the response from L2.1.

Expected Output
The response should now be:

- Easier to understand
- More detailed
- Relevant to engineering students
- Include practical applications such as:
    - Healthcare
    - Self-driving cars
    - Chatbots

Actual Output
Paste the AI-generated response here
Laboratory Task L2.3
Task Name
Optimize the Prompt Using Output Format and Constraints

Tool Used

- ChatGPT

Sample Prompt

Explain Artificial Intelligence for first-year CSE (AIML) students.
Instructions:
• Write within 150 words.
• Use simple English.
• Present the answer using headings and bullet points.
• Include one real-life example.
• End with a one-line conclusion.

Procedure

1. Open ChatGPT.
2. Copy and paste the optimized prompt.
3. Click **Send**.
4. Read the generated output.
5. Compare it with the outputs from L2.1 and L2.2.

Expected Output

The AI should produce:

- Well-structured answer
- Headings
- Bullet points
- Correct word limit
- Simple language
- Real-life example
- Conclusion

Actual Output

Paste the generated response here.

Laboratory Task L2.4

Task Name

Compare the Original and Optimized Prompts

Tool Used

- ChatGPT

Task/Prompt

Compare the responses generated in **L2.1**, **L2.2**, and **L2.3**.

Procedure

1. Review all three responses.
2. Observe the improvements made after prompt optimization.
3. Compare the responses using the given criteria.

4. Complete the comparison table.
5. Discuss your observations.

Comparison Criteria

- Clarity
- Relevance
- Completeness
- Readability
- Organization
- Output Format
- Usefulness for CSE Students

Comparison Table

Feature	Vague Prompt	Context Added	Fully Optimized Prompt
Clear Explanation			
Easy to Understand			
Well Organized			
Includes Examples			
Suitable for CSE Students			
Uses Headings			
Uses Bullet Points			
Overall Quality (1–5)			

Expected Output

Students should conclude that:

- The vague prompt generates a general response.
- Adding context improves relevance.
- Specifying output format and constraints produces the best-quality response.

Actual Output

Write your observations.

Laboratory Task L2.5

Task Name

Prepare a Prompt Optimization Report(give your own prompt considering sample report format)

Tool Used

- ChatGPT

Procedure

1. Review the outputs from L2.1 to L2.4.
2. Identify the improvements in each optimized prompt.
3. Write a short report.
4. Mention which prompt produced the best result.
5. Submit the report.

Sample Report Format

Title

Prompt Optimization Exercise

Objective

To study how prompt optimization improves AI-generated responses.

Observations

- The vague prompt produced a short and general response.
- Adding context generated a more relevant explanation.
- Specifying the output format produced a structured and informative answer.
- The optimized prompt was easier to understand and more suitable for first-year CSE and AIML students.

Conclusion

Prompt optimization improves the quality, clarity, relevance, and organization of AI-generated responses. Providing context, formatting instructions, and constraints helps AI produce more useful and accurate outputs.

Expected Output

Students should prepare a short report containing:

- Objective
- Observations
- Comparison
- Conclusion

Actual Output

Attach or write your report here.

Observation Table

Task No.	AI Tool Used	Type of Activity	Completed (✓/✗)
L2.1	ChatGPT	Vague Prompt	
L2.2	ChatGPT	Context-Based Prompt	
L2.3	ChatGPT	Optimized Prompt	
L2.4	ChatGPT	Prompt Comparison	
L2.5	ChatGPT	Prompt Optimization Report	

Result

The experiment was successfully completed. Students learned how prompt optimization techniques such as adding context, specifying output formats, and applying constraints improve the quality, clarity, and usefulness of AI-generated responses. They also observed that well-structured prompts produce more accurate and organized outputs than vague prompts.

Learning Outcomes

After completing this experiment, students will be able to:

1. Understand the importance of prompt optimization.
2. Differentiate between vague and optimized prompts.
3. Apply context and constraints to improve AI responses.
4. Generate structured and relevant outputs using AI tools.

5. Evaluate AI responses based on clarity, accuracy, and organization.
6. Use prompt engineering techniques for academic and technical applications.
7. Develop better interaction skills with Generative AI tools.

Viva-Voce Questions

1. What is Prompt Optimization?
2. Why are vague prompts less effective?
3. What is meant by adding context to a prompt?
4. How do constraints improve AI responses?
5. Why is specifying an output format useful?
6. What are the characteristics of a good prompt?
7. Which prompt produced the best response in this experiment?
8. How can prompt optimization help CSE and AIML students?
9. Name two AI tools used for prompt engineering.
10. Why is prompt engineering important in Generative AI?

Assignment

Topic: Prompt Optimization for Programming Concepts

1. Write a **vague prompt** to explain **Python Functions**.
2. Improve the prompt by adding context for **first-year CSE students**.
3. Optimize the prompt further by specifying:
    - Word limit (150 words)
    - Bullet points
    - One Python code example
    - Simple language
4. Generate responses for all three prompts using **ChatGPT or Gemini**.
5. Compare the responses and write a **150-word report** explaining how prompt optimization improved the quality of the output.

Experiment 3:Building a Content Automation Workflow using n8n

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Aim

To understand the basic concepts of workflow automation using **n8n** and learn how AI can automate content generation and simple publishing tasks.

Objective

After completing this experiment, students will be able to:

- Understand the concept of workflow automation.
- Explore the n8n interface and identify its main components.
- Create a simple workflow using nodes and connections.
- Design an AI-assisted content generation workflow.
- Understand how workflows can automate repetitive tasks using AI.

Tools Required

- n8n (Cloud Version: https://n8n.io)
- ChatGPT (https://chatgpt.com) or Google Gemini (https://gemini.google.com)
- Internet Connection

- Web Browser (Google Chrome / Microsoft Edge)

Note: If installing n8n is not feasible, students may use the **n8n Cloud Trial**, watch the official n8n demo, or draw the workflow with screenshots from tutorials.

General Procedure

Step 1: Open Google Chrome or Microsoft Edge.
Step 2: Visit **https://n8n.io** and sign in or create a free account.
Step 3: Create a new workflow.
Step 4: Add the required nodes and connect them.
Step 5: Execute the workflow, observe the output, and record your observations.

Introduction

Workflow Automation is the process of performing repetitive tasks automatically using software. Instead of manually completing each step, workflows connect multiple tasks together so that they execute in sequence.

n8n is a no-code/low-code workflow automation platform that allows users to connect applications and AI tools using graphical nodes.

A workflow consists of:

- **Trigger Node** – Starts the workflow.
- **Processing Node** – Performs an action (e.g., AI text generation).
- **Output Node** – Displays, stores, or sends the generated result.

Simple Workflow

Manual Trigger
↓
AI Tool (ChatGPT/Gemini)
↓
Generated Content

Students are **not expected to write code**. The focus is on understanding how workflows are designed and executed.

Laboratory Task L3.1

Task Name

Explore the n8n Dashboard

Tool Used

- n8n Cloud

Procedure

1. Open **https://n8n.io**.
2. Create a free account or sign in.
3. Open the **Dashboard**.
4. Explore the following sections:
    - Workflows
    - Templates
    - Credentials
    - Executions

    - Nodes
5. Record your observations.

Expected Output
Students should identify the following components:

- Dashboard
- Create Workflow Button
- Workflow Canvas
- Nodes Panel
- Execute Workflow Button

Students should understand the purpose of each section.

Actual Output
Write the purpose of the following n8n components.

Component	Purpose
Dashboard	
Workflow	
Nodes	
Execute Button	
Templates	

Laboratory Task L3.2
Task Name
Create a Simple Workflow (Trigger → AI Node)

Tool Used

- n8n
- ChatGPT or Gemini

Sample AI Prompt
Explain the importance of Artificial Intelligence for first-year CSE students in five simple points.

Procedure

1. Open a new workflow in n8n.
2. Add a **Manual Trigger** node.
3. Add an **AI Node** (or simulate by connecting to ChatGPT/Gemini manually if an AI node is unavailable).
4. Enter the sample prompt.
5. Execute the workflow and observe the generated output.

Workflow Diagram
Manual Trigger
↓
AI Text Generator
↓

Generated Answer

Expected Output
The workflow should generate five simple points explaining the importance of Artificial Intelligence for CSE students.

Actual Output
Paste or attach the AI-generated response.

Laboratory Task L3.3
Task Name
Design an AI-Based Social Media Content Workflow

Tool Used

- n8n
- ChatGPT or Gemini

Sample Prompt
Generate a LinkedIn post (100 words) announcing the inauguration of the Artificial Intelligence Club in our engineering college. Use a professional tone and include three relevant hashtags.

Procedure

1. Open the workflow created in L3.2.
2. Add an AI node or simulate AI content generation.
3. Enter the sample prompt.
4. Execute the workflow.
5. Review the generated social media post.

Workflow Diagram
Manual Trigger
↓
ChatGPT/Gemini
↓
LinkedIn Post

Expected Output
The workflow should generate a professional LinkedIn post containing:

- Event announcement
- Short description
- Three hashtags
- Professional writing style

Actual Output

Paste the generated post.

Laboratory Task L3.4
Task Name
Extend the Workflow by Adding an Output Node

Tool Used
- n8n

Task
Extend the previous workflow by adding another node to save or simulate publishing the generated content.

Procedure
1. Open the workflow created in L3.3.
2. Add an additional output node (e.g., Google Docs, Email, Notion, or a simulated output node).
3. Connect it to the AI node.
4. Execute the workflow.
5. Observe how the content flows through each node.

Workflow Diagram
Manual Trigger
↓
ChatGPT/Gemini
↓
Save Output
↓
Simulated Publishing

Expected Output
The workflow should demonstrate how AI-generated content can be automatically:
- Saved
- Sent
- Displayed
- Simulated for publishing

Students should understand that workflow automation reduces manual effort.

Actual Output
Draw or paste your workflow.

Laboratory Task L3.5
Task Name
Draw and Explain the Workflow

Tool Used

- n8n
- Drawing Tool / Notebook

Procedure

1. Observe the workflow created in the previous tasks.
2. Draw the workflow neatly on paper or using a drawing tool.
3. Label each node.
4. Write the purpose of every node.
5. Submit the completed workflow diagram.

Sample Workflow

Start



Manual Trigger



ChatGPT / Gemini



Generate Social Media Content



Save Output



End

Explain the Role of Each Node

Node	Purpose
Manual Trigger	Starts the workflow
AI Node	Generates the required content
Save Output	Stores the generated content
End	Workflow completion

Expected Output

Students should submit:

- Workflow diagram
- Explanation of each node
- Understanding of the workflow sequence

Actual Output

Attach the workflow diagram here.

Observation Table

Task No.	Tool Used	Activity	Completed (✓/✗)
L3.1	n8n	Dashboard Exploration	
L3.2	n8n + ChatGPT/Gemini	Basic Workflow	
L3.3	n8n + ChatGPT/Gemini	AI Content Generation	
L3.4	n8n	Workflow Extension	
L3.5	n8n	Workflow Diagram & Explanation	

Result

The experiment was successfully completed. Students explored the n8n platform, identified its key components, created a simple AI-assisted workflow, extended it with an output node, and understood how workflow automation can simplify repetitive tasks such as AI-based content generation and publishing.

Learning Outcomes

After completing this experiment, students will be able to:

1. Explain the concept of workflow automation.
2. Identify the major components of the n8n interface.
3. Create a simple workflow using nodes and connections.
4. Understand how AI tools can be integrated into automated workflows.
5. Design a basic AI-powered content generation workflow.
6. Explain the role of trigger, processing, and output nodes.
7. Recognize practical applications of workflow automation in software engineering, AI, and digital content creation.

Viva-Voce Questions

1. What is workflow automation?
2. What is n8n?
3. What is a node in n8n?
4. What is the purpose of a Trigger node?
5. Which node is responsible for AI content generation?
6. Why are connections required between nodes?
7. What is the advantage of automating repetitive tasks?
8. Name two real-world applications of n8n.
9. Can AI tools like ChatGPT or Gemini be integrated into workflows? Explain.
10. How can workflow automation benefit CSE and AIML professionals?

Assignment

Topic: AI Workflow for Student Notifications

1. Design a workflow to automate **college event notifications**.
2. The workflow should include:
    - **Manual Trigger**
    - **AI Tool (ChatGPT/Gemini)** to generate a notification
    - **Save Output** node
    - **Email or Messaging** node (or simulate sending)
3. Draw the workflow diagram.
4. Explain the function of each node in **2–3 lines**.

5. Write a **100–150 word note** on how workflow automation can save time and improve productivity in educational institutions.

Instructor Note: If students cannot install or access n8n, they may complete this experiment using the **n8n Cloud interface**, screenshots from official tutorials, or neatly drawn workflow diagrams. The objective is to understand workflow design rather than deployment or programming.

Experiment 4:API and RSS Feed Integration for Real-Time Summaries

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Aim

To understand how RSS Feeds, APIs, and AI tools work together to collect, process, and summarize real-time information.

Objective

After completing this experiment, students will be able to:

- Understand the concept of RSS Feeds and APIs.
- Explore public RSS feeds to collect the latest news.
- Use Generative AI tools to summarize news articles.
- Understand the workflow of real-time information retrieval.
- Learn the role of SerpAPI and AI in automating information gathering.

Tools Required

- BBC News RSS Feed (https://feeds.bbci.co.uk/news/rss.xml) or The Hindu RSS Feed (https://www.thehindu.com/feeder/default.rss)
- ChatGPT (https://chatgpt.com) or Google Gemini (https://gemini.google.com)
- Web Browser (Google Chrome / Microsoft Edge)
- Internet Connection

Note: Students are **not required to write any code**. The experiment focuses on understanding the concepts of RSS feeds, APIs, AI summarization, and workflow automation.

General Procedure

Step 1: Open Google Chrome or Microsoft Edge.
Step 2: Visit a public RSS Feed website (BBC News or The Hindu RSS).
Step 3: Select a recent news article and copy its content.
Step 4: Open ChatGPT or Gemini and paste the article with the summarization prompt.
Step 5: Observe the generated summary and record your observations.

Introduction

Every day, thousands of news articles are published on different websites. Reading all of them manually is difficult. Technologies like **RSS Feeds**, **APIs**, and **Generative AI** help users receive the latest information quickly and efficiently.

- **RSS Feed (Really Simple Syndication):** Automatically delivers the latest updates from websites in a standardized format.
- **API (Application Programming Interface):** Allows different software applications to exchange information.

- **Generative AI:** Reads and summarizes large amounts of text into short, meaningful content.
- **SerpAPI:** Provides structured search engine results that can be integrated into AI workflows for real-time information retrieval.

These technologies work together to automate information collection and summarization.

Laboratory Task L4.1
Task Name
Explore a Public RSS Feed

Tool Used

- BBC News RSS Feed or The Hindu RSS Feed

Procedure

1. Open Google Chrome or Microsoft Edge.
2. Visit the BBC News RSS Feed or The Hindu RSS Feed.
3. Browse the latest news headlines.
4. Select five recent headlines.
5. Record the headlines in your observation sheet.

Expected Output
Students should identify the latest news headlines published through the RSS feed.
Example:

S.No	News Headline
1	AI Revolution in Education
2	New Technology Innovations
3	Climate Change Updates
4	Space Exploration Mission
5	Cybersecurity Developments

Actual Output
Write the latest five news headlines.

S.No	News Headline
1	
2	
3	
4	
5	

Laboratory Task L4.2
Task Name
Select and Copy a News Article

Tool Used

- BBC News or The Hindu Website

Procedure

1. Select one interesting news article from the RSS feed.
2. Open the complete article.
3. Read the article carefully.
4. Copy the main content (approximately 200–300 words).
5. Save the copied content for summarization.

Suggested Topic (CSE & AIML Students)

Choose a technology-related article such as:

- Artificial Intelligence
- Machine Learning
- Robotics
- Cyber Security
- Cloud Computing
- Data Science
- Space Technology

Expected Output

Students should collect one complete news article suitable for summarization.

Actual Output

Mention the article title.

Article Title:

Source:

Laboratory Task L4.3

Task Name

Summarize the News Article Using AI

Tool Used

- ChatGPT or Google Gemini

Sample Prompt

Summarize the following news article in about 100 words.

Instructions:

• Use simple English.

• Highlight the main points.

• Write in paragraph form.

• Suitable for first-year students.

(Paste the copied news article here.)

Procedure

1. Open ChatGPT or Gemini.

2. Sign in using your account.
3. Paste the copied news article into the prompt box.
4. Enter the sample prompt.
5. Click **Send** and review the generated summary.

Expected Output
The AI should generate:

- A concise summary (approximately 100 words).
- Simple and easy-to-understand language.
- Important facts from the article.
- Well-organized paragraph.

Actual Output
Paste or attach the AI-generated summary.

Laboratory Task L4.4
Task Name
Draw the RSS Feed to AI Workflow

Tool Used

- Notebook / Drawing Tool (MS Paint, PowerPoint, Draw.io, etc.)

Procedure

1. Observe the process followed in Tasks L4.1 to L4.3.
2. Draw the workflow diagram.
3. Label each component clearly.
4. Write one line describing the function of each step.
5. Submit the completed workflow.

Sample Workflow Diagram
RSS Feed

Latest News Articles

ChatGPT / Gemini

AI Summary

Student / User

Role of Each Component

Component	Function
RSS Feed	Provides the latest news updates
News Article	Input data for summarization
ChatGPT / Gemini	Generates a concise summary
User	Reads and understands the summarized information

Expected Output
Students should draw the workflow correctly and explain the role of each component.

Actual Output
Attach or draw the workflow diagram

Laboratory Task L4.5
Task Name
Write a Note on RSS Feeds, APIs, AI, and SerpAPI

Tool Used

- ChatGPT or Gemini (Optional)
- Notebook / MS Word

Procedure

1. Read about RSS Feeds, APIs, AI, and SerpAPI.
2. Write a short note explaining the purpose of each technology.
3. Explain how they work together for real-time information retrieval.
4. Mention one practical application.
5. Submit your notes.

Sample Prompt
Explain the role of RSS Feeds, APIs, Artificial Intelligence, and SerpAPI in real-time information retrieval using simple language suitable for first-year CSE and AIML students.

Expected Output
Students should explain:
RSS Feed
Provides automatic updates whenever new content is published.
API
Allows different software applications to exchange information automatically.
Artificial Intelligence
Analyzes and summarizes collected information quickly and accurately.
SerpAPI
Retrieves structured search engine results that can be used in AI-powered workflows for collecting relevant information.
Real-World Applications

- News aggregation
- Research assistance
- Stock market updates

- Weather alerts
- Educational content summarization

Actual Output
Write your explanation here.

Observation Table

Task No.	Tool Used	Activity	Completed (✓/✗)
L4.1	BBC/The Hindu RSS	RSS Feed Exploration	
L4.2	News Website	Article Collection	
L4.3	ChatGPT/Gemini	AI-Based Summarization	
L4.4	Drawing Tool	Workflow Design	
L4.5	ChatGPT/Gemini	Concept Explanation	

Result
The experiment was successfully completed. Students explored public RSS feeds, collected a recent news article, generated a concise AI-based summary, designed a simple workflow illustrating RSS Feed to AI summarization, and understood the roles of RSS Feeds, APIs, AI, and SerpAPI in real-time information retrieval.

Laboratory Task L4.6(additional)
Task Name
Explore and Work with SerpAPI
Tool Used
- SerpAPI (https://serpapi.com)
- Web Browser (Google Chrome / Microsoft Edge)
- ChatGPT or Google Gemini (Optional)

Note: Students are **not required to write code or use an API key**. This experiment focuses on understanding the concept and exploring the SerpAPI interface.

Sample Search Query
Artificial Intelligence in Education
or
Latest Machine Learning News
or
Applications of Artificial Intelligence in Healthcare

Procedure
Step 1: Open Google Chrome or Microsoft Edge and visit **https://serpapi.com**.
Step 2: Create a free account or sign in (optional for exploring the website and documentation).
Step 3: Navigate to the **Playground**, **Demo**, or **Documentation** section and observe how SerpAPI retrieves search engine results.
Step 4: Enter a sample search query (or examine the example queries provided on the website) and observe the returned search results such as titles, snippets, and URLs.
Step 5: Record your observations and explain how SerpAPI can be integrated with AI tools like ChatGPT or Gemini for real-time information retrieval and summarization.

Sample Activity
Search for:
Latest Artificial Intelligence Trends
Observe the following:

- Top search result
- Website name
- Short description (snippet)
- Related search results

(Optional) Copy one search result summary and use ChatGPT or Gemini to generate a concise summary.

Expected Output
Students should be able to:

- Explain the purpose of SerpAPI.
- Observe how search results are retrieved in a structured format.
- Identify titles, URLs, and snippets from search results.
- Understand that SerpAPI acts as a bridge between search engines and AI applications.
- Describe how AI can summarize the retrieved information.

Actual Output
Search Query

Top Search Result
Website Name
Summary / Observation

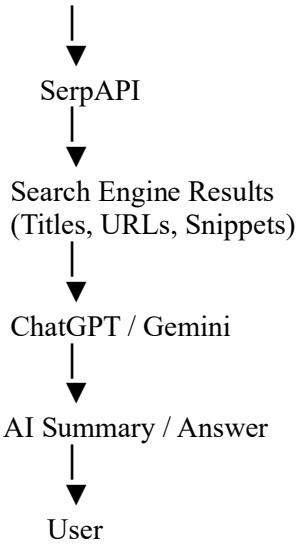
Observation Table

Feature	Observation
Search Query	
Number of Results Observed	
Website Names	
Information Retrieved	
Can AI Summarize the Results?	Yes / No
Real-Time Information Available	Yes / No

Expected Learning
After completing this task, students will understand that:

- **SerpAPI** retrieves real-time search results from search engines.
- It provides structured information such as **titles, URLs, snippets, images, and news**.
- AI tools like **ChatGPT** or **Gemini** can use this information to generate summaries or answer questions.
- SerpAPI is commonly used in **AI agents**, **chatbots**, **workflow automation**, **research assistants**, and **news aggregation systems**.

Real-Time Information Retrieval Workflow
User Search Query



Conclusion

SerpAPI simplifies the process of retrieving real-time information from search engines by providing structured search results. When integrated with AI tools such as ChatGPT or Gemini, it enables intelligent applications to collect, summarize, and present up-to-date information efficiently. This makes SerpAPI a valuable component in AI-powered workflows, research assistants, and automation systems.

Learning Outcomes

After completing this experiment, students will be able to:

1. Explain the purpose of RSS Feeds and APIs.
2. Explore public RSS feeds to access real-time information.
3. Use AI tools to summarize lengthy articles.
4. Draw a simple workflow representing AI-assisted information processing.
5. Describe the role of SerpAPI in information retrieval.
6. Understand how AI simplifies information gathering and summarization.
7. Recognize real-world applications of AI, APIs, and RSS Feeds in education, research, and technology.

Viva-Voce Questions

1. What is an RSS Feed?
2. What is the purpose of an API?
3. What does RSS stand for?
4. How does ChatGPT help in summarization?
5. What is SerpAPI?
6. Why are RSS Feeds useful?
7. Name two websites that provide RSS Feeds.
8. What is real-time information retrieval?
9. How do APIs and AI work together?
10. Mention two real-world applications of AI-based summarization.

Assignment

Topic: AI-Based News Summarization

1. Visit a public RSS Feed such as **BBC News** or **The Hindu**.
2. Select a technology-related news article (e.g., Artificial Intelligence, Machine Learning, Cybersecurity, Robotics, or Data Science).
3. Copy the article and summarize it into **100 words** using **ChatGPT or Gemini**.
4. Draw the workflow: **RSS Feed → AI Tool → Summary → User**.
5. Write a **150-word note** explaining how **RSS Feeds, APIs, AI, and SerpAPI** work together to provide real-time information and discuss one practical application in the field of **CSE or AIML**.